

Week 1: The Pile Game

Bulgarian solitaire · Facilitator guide for the three table adults

The theorem of the day. Counters sit in piles. A move takes one counter from every pile and makes a new pile out of them. If the number of counters is a *staircase number* $1 + 2 + \dots + k$, then *every* start ends stuck at the staircase $(k, k - 1, \dots, 1)$; for any other number of counters the game never gets stuck and falls into a loop instead. The children will discover this, and the older ones will prove the parts of it that fit in an hour.

1 At a glance

Time	Where	What
0:00–0:04	tables	Free handling: cups of counters and cards on each table; children make piles and count them.
0:04–0:12	whole group	Launch (§4): the ten children <i>are</i> the counters; two groups of five end stuck at 4 3 2 1. The organizer records the game on the board and poses the question.
0:12–0:50	tables	Table work (§5, §6, §7), each table with its own pages and its own adult.
0:50–0:58	whole group	Share (§8): one discovery per table, youngest first.
0:58–1:00	whole group	Closing: the organizer names the theorem and gives the take-home question.

Who sits where. The K–1 run sheet (§5) is written so that the parent can run that table without any mathematical background; the grades 2–3 table needs someone comfortable with the “why is the staircase stuck” argument; the grades 4–5 table needs a mathematician, because the children will be proving things. Suggested: parent at K–1, second mathematician at 2–3, organizer at 4–5. Swap freely.

Reading order for a busy adult. §2 (preparation), §3 up to Fact 4 (the mathematics everyone needs), then only your own table’s section. Appendix A has every number you might be asked for.

2 Materials and preparation

- **Counters:** about 50 per table in a cup or bowl (grades 4–5 will want all 50 for the 15-counter challenge; at K–1 give each child 15–20 of their own on a paper plate and keep the rest for the shared games). Any kind; one colour is fine. A handful of a *second* kind (coins, buttons, a different colour) is useful at the 2–3 and 4–5 tables for marking “extra” counters (see §6 step 4 and §7 step 4). Not essential.
- **Playing cards:** one deck per table, optional. Children can lay a card next to each pile showing its size, which speeds up recording and makes “the sizes are what matter” visible. K–1 p. 6 has a prompt for this.
- **Student pages,** printed single-sided (children draw on them): K–1, 6 pages, one set per child (3 sets + 1 spare); grades 2–3, 6 pages (4 sets + 1 spare); grades 4–5, 6 pages (3 sets + 1 spare). Pencils, and a few spare sheets of blank paper per table.
- **For the launch:** a whiteboard or large paper visible from the floor, a marker, and a clear space where ten children can stand in groups. A chair or a taped square marks where each “new group” forms. If there is no floor space, use ten large counters (or the children’s name cards) on a table with everyone gathered round.

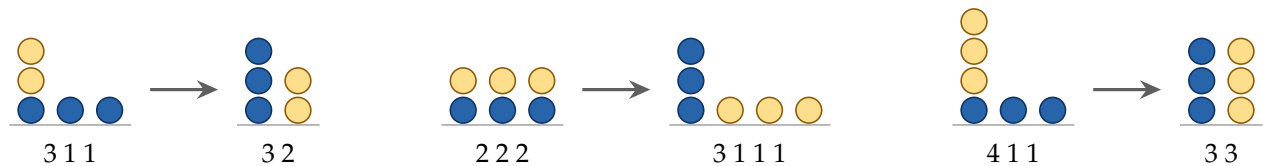
- **For the share:** the 2–3 table will hold up their map of six (p. 3 of their pages); the K–1 table shows staircases built from counters, so leave their staircases on the table at the end.
- **Print this guide once per adult.** Each adult only needs to read their own table’s section in detail, but the launch and share involve everyone.

Before the session (10 minutes). Play the game yourself, once with 6 counters from 4 1 1 and once with 8 counters from 8, so that you have felt a pile disappear and a loop begin. At the K–1 table, the adult reads the pages aloud; skim them beforehand.

3 The mathematics you need

3.1 Rule, notation, words

There are n counters in piles. A **position** is the list of pile sizes, written biggest first: **4 1 1**. (Where the piles sit does not matter; only the sizes do. A mathematician would say a position is a *partition* of n .) A **move**: take one counter from every pile; those counters together form a new pile; a pile that had one counter has disappeared. So if there were m piles, the new pile has m counters and every old pile is one smaller.



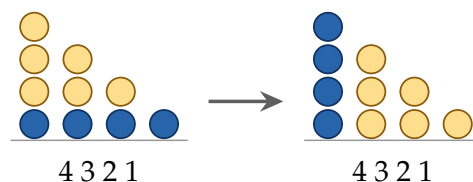
Words we use with the children (with the grown-up word in brackets): **stuck** (fixed point) when the move gives the same position back; **staircase** $k, k - 1, \dots, 1$ and **staircase numbers** $T_k = 1 + 2 + \dots + k = k(k + 1)/2$, that is 1, 3, 6, 10, 15, 21, 28, ... (triangular numbers); **loop** (cycle) when positions come back round and round; **road** (trajectory) for the sequence of positions from a start, and its **length** in moves until the game is stuck or the loop begins; **map** (state diagram) with an arrow from every position to the next; **Garden of Eden** for a position that no move can produce.

3.2 What happens, in one paragraph

For 1, 3, 6, 10, 15, ... counters (staircase numbers) every start ends stuck at the staircase. For every other number of counters nothing is ever stuck; every start ends in a loop. With 7 counters there is one loop of four positions; with 8 there are two loops (one of four positions, one of two); with 9 one loop of four; with 11 one loop of five; with 12 two loops of five. Appendix A lists everything up to 15 counters.

3.3 The facts, with the arguments the children can find

Fact 1. The staircase is stuck. Lay the piles out as columns, tallest on the left. Taking one counter from every pile is taking the *bottom row*. The bottom row of the staircase $k, k - 1, \dots, 1$ has k counters (one per pile), so it stands up as a new pile of k ; the old columns shrink to $k - 1, \dots, 1, 0$. Put the new column at the left: the same staircase.

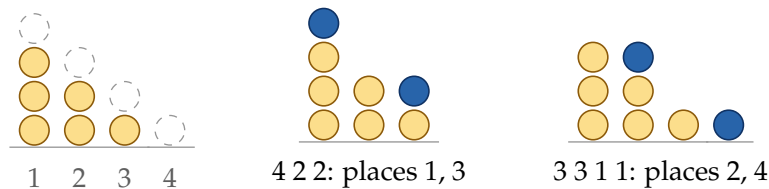


Fact 2. Nothing else is stuck. Suppose a position with m piles, sizes $p_1 \geq p_2 \geq \dots \geq p_m$, is stuck. After the move there is a new pile of m counters and every old pile is one smaller (the old piles of size 1 have

gone). The biggest pile after the move is either the new pile or $p_1 - 1$; it must equal p_1 , and $p_1 - 1 \neq p_1$, so the biggest pile is the new one: $p_1 = m$. Now set that pile aside on both sides. What is left after the move is $p_1 - 1, p_2 - 1, \dots$ (zeros removed); what is left of the original is $p_2, p_3, \dots, p_m$. These are the same collection of piles, so compare them from the biggest down: $p_1 - 1 = p_2$, then $p_2 - 1 = p_3$, and so on. Each pile is one smaller than the one before, starting from m , and there are m piles: the position is $m, m - 1, \dots, 1$, a staircase. *Consequence:* a number that is not a staircase number (like 8) can never get stuck, because there is no staircase to get stuck at.

Fact 3. Every game eventually repeats. There are finitely many positions ($p(n)$ of them: 11 for six counters, 42 for ten), and the next position depends only on the current one. After $p(n)$ moves you have seen $p(n) + 1$ positions, so some position appeared twice, and from its second appearance the game repeats what it did after the first. So every road ends in a loop—possibly a loop of length one, which is what “stuck” means.

Fact 4. The staircase plus extras. Take $n = T_k + r$ counters with $0 \leq r \leq k$. Start from the staircase $k, \dots, 1$ and add r extra counters, at most one in each of $k + 1$ places: place $j \leq k$ means “on top of pile j ”; place $k + 1$ means “a new pile of one”. For 7 counters ($k = 3, r = 1$) the four such positions are **4 2 1**, **3 3 1**, **3 2 2**, **3 2 1 1**. *Claim:* a move sends such a position to another one, moving every extra one place along (place $j \rightarrow j + 1$, and place $k + 1 \rightarrow$ place 1). Because: the bottom row has k counters, plus one more if place $k + 1$ is occupied—and that lone counter is exactly the pile that vanishes. The bottom row stands up as the new first column, so every old column shifts one place right, carrying its extra with it, and the new first column carries an extra precisely when place $k + 1$ was occupied. So these $\binom{k+1}{r}$ positions form loops, and every loop length divides $k + 1$. For 8 counters ($r = 2$): extras in places $\{1, 2\} \rightarrow \{2, 3\} \rightarrow \{3, 4\} \rightarrow \{4, 1\} \rightarrow \{1, 2\}$ gives the loop **4 3 1** $\rightarrow$ **3 3 2** $\rightarrow$ **3 2 2 1** $\rightarrow$ **4 2 1 1**, and $\{1, 3\} \rightarrow \{2, 4\} \rightarrow \{3, 1\}$ gives the two-loop **4 2 2** $\rightarrow$ **3 3 1 1**.



Fact 5 (Brandt’s theorem, 1982). Every position in a loop is of the form in Fact 4. Hence for a staircase number of counters ($r = 0$) the only loop is the staircase itself, and by Fact 3 every start reaches it; for $T_k < n < T_{k+1}$ every start ends in one of the Fact 4 loops. The proof is a few pages of careful bookkeeping and is not for the session; the children verify the statement experimentally for 7, 8, 9, 11 and 12 counters.

Fact 6 (how long the road is). For T_k counters the road to the staircase has at most $k^2 - k$ moves (Igusa 1985; Étienne 1991), and for every $k \leq 6$ some start needs exactly that many (Appendix A). Records: 2 moves for 3 counters (from **1 1 1**), 6 for 6 counters (from **2 2 1 1**), 12 for 10 counters (from **3 3 2 1 1**, **3 2 2 2 1** or **3 2 2 1 1 1**), 20 for 15 counters, 30 for 21.

Fact 7 (going backwards; Gardens of Eden). Suppose a position with p piles was produced by a move. Its new pile has as many counters as there were piles before, and every other pile came from an old pile of size at least 2; so the number of other piles is at most the size of the new pile, and $p \leq (\text{new pile}) + 1 \leq (\text{biggest pile}) + 1$. Conversely, if some pile has size $s \geq p - 1$, here is a parent: remove that pile, add one counter to each of the other $p - 1$ piles, and add $s - (p - 1)$ piles of one. (Check: the parent has s piles, so its move makes a new pile of s , shrinks the others back, and the ones vanish.) So a position is a Garden of Eden exactly when its biggest pile is smaller than $p - 1$, i.e. it has at least two more piles than its biggest pile has counters; and the number of parents of a position equals

the number of *different* pile sizes s with $s \geq p - 1$. Gardens of Eden with 6 counters: **2 2 1 1**, **2 1 1 1 1**, **1 1 1 1 1 1**. With 7: **3 1 1 1 1**, **2 2 2 1**, **2 2 1 1 1**, **2 1 1 1 1 1**, **1 1 1 1 1 1 1**. The position **3 2** has two parents (**4 1** and **3 1 1**); **4 3 2** has three (**5 4**, **5 3 1**, **4 3 1 1**); the smallest position with four parents is **6 5 4 3**, with 18 counters.

3.4 For the mathematicians

A position is a partition λ of n ; the move removes the first column of the Young diagram and adds it as a new row (or vice versa), which is why the literature calls the map a *shift*. Brandt proved that the cycle partitions for $T_k \leq n < T_{k+1}$, $n = T_k + r$, are exactly the $(k + \varepsilon_1, k - 1 + \varepsilon_2, \dots, 1 + \varepsilon_k, \varepsilon_{k+1})$ with $\varepsilon_i \in \{0, 1\}$ and $\sum \varepsilon_i = r$; the move rotates the ε -word cyclically, so cycles correspond to binary necklaces of length $k + 1$ with r ones, and cycle lengths divide $k + 1$ (Fact 4 above is the easy half). For $k + 1$ prime and $0 < r < k + 1$ every cycle therefore has length exactly $k + 1$: with 11, 12, 13, 14 counters all loops have length 5. Igusa and Étienne proved the bound $k^2 - k$ on the transient for $n = T_k$; Griggs and Ho (1998) studied the cycles for general n and the lengths of transients; Akin and Davis (1985) is an early thorough analysis, including positions with no predecessor. Gardner's August 1983 *Scientific American* column made the game famous. See the references at the end.

4 The whole-group launch (0:04–0:12)

The organizer runs this; the other two adults stand by the “new group” spot and help children count. The ten children are the counters. It is deliberate that 10 is a staircase number: whatever they do, they will end at **4 3 2 1**.

*“Everybody stand up—today you are the counters. Make two groups of five.” [Write **5 5** on the board.] “Here is the only rule of the game. Every group sends one person to the new-group spot. Those people are a new group. Go!”*
*[Count aloud with them.] “The old groups have 4 and 4, the new group has 2.” [Write **4 4 2**.]*
*“Again: every group sends one person.” [**3 3 3 1**. Point at the group of one:] “Watch this group of one. When they send their person, nobody is left—the group is gone.”*
*“Again.” [**4 2 2 2**—the new group has four because there were four groups.]*
*“Again.” [**4 3 1 1 1**.] “Again.” [**5 3 2**: three groups of one vanish at once.] “Again.” [**4 3 2 1**.]*
*“Again.” [**4 3 2 1**.] “Look at the board. Nothing changed! One more time to be sure.” [**4 3 2 1**.] “It’s stuck.”*

That is six moves plus two checks, about four minutes. Then pose the question, and send the tables off:

“We started with 5 and 5 and got stuck at 4, 3, 2, 1. Was that luck? Would a different start get stuck at the same place? What if there were eight of us instead of ten? At your tables you have counters instead of people. Go and find out.”

Launch notes

- Say the “group of one disappears” rule *before* it happens (it first happens at **3 3 3 1**).
- Which person leaves a group does not matter—say so if a child asks; it is the first step towards “only the sizes count”.
- Keep the board record; that column of numbers is the notation the tables will use. Write the sizes biggest first.
- If the room has no floor space: ten big counters on a table with the children standing round, same script. From **5 5** the road is **5 5** → **4 4 2** → **3 3 3 1** → **4 2 2 2** → **4 3 1 1 1** → **5 3 2** → **4 3 2 1**.

- If a child is absent (nine counters) the game loops instead: **5 4** → **4 3 2** → **3 3 2 1** → **4 2 2 1** → **4 3 1 1** → **4 3 2**. That is a fine launch too—*“it never stops! Is that because of the number of people?”*—and the K–1 page 6 and the 2–3 page 5 then feel familiar. With eleven people: **6 5** → **5 4 2** → **4 3 3 1** → **4 3 2 2** → **4 3 2 1 1** → **5 3 2 1** → **4 4 2 1** → **4 3 3 1**, a loop of five.

5 K–1 table (two kindergartners, one first grader)

The adult reads every page aloud; the children play with real counters and draw circles. The goal for this table is modest and real: know the rule, feel “stuck”, see that six counters always end at the same staircase, meet the staircase numbers, and see that four counters go round and round. Each child has their own 15–20 counters on a paper plate; the rest of the table’s cup is for the ten- and fifteen-counter games at the end. Keep the children’s counters apart.

Order of work, with times

Minutes	Page	What happens
0:12–0:17	p. 1	The rule with counters: 3 1 1 → 3 2, then 2 2 2 → 3 1 1 1. Draw the second result.
0:17–0:23	p. 2	Three counters from 1 1 1 and from 3. Both get stuck at 2 1. The word “stuck”.
0:23–0:33	p. 3	Six counters from any start. Everyone ends at 3 2 1. “Is that a coincidence?” Try another start.
0:33–0:39	p. 4	Staircases: count 10, build 5 4 3 2 1 (15). Eight counters cannot make one.
0:39–0:46	p. 5	Four counters: never stuck; the merry-go-round 3 1 → 2 2 → 2 1 1 → 3 1.
0:46–0:50	p. 6	Whatever there is time for: ten counters together, five counters, fifteen, cards.

How to run a move with five-year-olds

Ask the child to touch each pile and slide one counter out of it into a line in front of the piles (*“one from this pile, one from this pile...”*). Then the line is pushed together: *“this is the new pile”*. Then ask *“are any piles empty now? They’re gone.”* Laying the piles out as columns (a straight line of counters going away from the child) makes “one from every pile” the same as “the row nearest you”, which is the picture on the pages. Drawing: circles in a box, one column per pile. If drawing is slow, the adult draws and the child says the numbers.

Answers, page by page

Page 1: the rule

3 1 1 → **3 2**. **2 2 2** → **3 1 1 1**: four piles; the new pile is the one with three; no pile disappeared this time, every pile just got smaller.

Page 2: three counters

1 1 1 → **3** → **2 1** → **2 1**: stuck after two moves. **3** → **2 1** → **2 1**: stuck after one move. Same stuck piles both times. There is only one other start, **2 1** itself, which is stuck from the beginning—a nice thing for a first grader to notice.

Page 3: six counters

Every one of the eleven starts ends at **3 2 1**. Longest road: **2 2 1 1** → **4 1 1** → **3 3** → **2 2 2** → **3 1 1 1** → **4 2** → **3 2 1**, six moves; the strip on the page has room for seven. Shortest: **4 2** takes one move. From **6**: **5 1**, **4 2**, **3 2 1**. From **1 1 1 1 1 1**: **6**, then as before. From **3 3**: **2 2 2**, **3 1 1 1**, **4 2**, **3 2 1**. The full map is in Appendix A. Expected answer to “did everyone get stuck at the same piles?”: yes—and that is exactly the theorem.

Page 4: staircases

4 3 2 1 has 10 counters. The next staircase, **5 4 3 2 1**, has 15. The staircase numbers go 1, 3, 6, 10, 15: each time you add the next number (add 2, add 3, add 4, add 5). Eight counters: a staircase with three steps uses 6 and leaves 2, which is not enough for a fourth step of 4; with four steps you need 10. So 8 cannot be a staircase.

Page 5: four counters

1 1 1 1 → **4** → **3 1** → **2 2** → **2 1 1** → **3 1** → **2 2** → ... It never gets stuck (NO). The merry-go-round is **3 1** → **2 2** → **2 1 1** → back to **3 1**. Any start with four counters lands on it: **4** goes to **3 1** at once.

Page 6: more to try

Ten counters: stuck at **4 3 2 1**, a staircase, just like the people. From **5 5** it takes 6 moves; the longest road is 12 moves (from **3 3 2 1 1**); from **10** it is 6. **Five counters:** a merry-go-round of three: **3 2** → **2 2 1** → **3 1 1** → **3 2**. From **5**: **4 1**, then **3 2**. **Fifteen counters:** **5 4 3 2 1**, five piles. **Cards:** yes, after every move the sizes change, so the cards must change; a card is a way of writing the position.

If the K–1 table stalls

- A child takes all the counters from one pile instead of one from each: go back to the line-in-front method, one finger on each pile.
- A child does not notice that the position is stuck: “Do the move. Now look—did anything change? Try again.” Let them do it three times; the repetition is the point.
- A child loses track during the six-counter game: it does not matter; start again from any position. The ending is the same (that is the theorem).
- The word “staircase” means nothing to them: arrange **3 2 1** as columns and walk two fingers down the steps.
- A kindergartner cannot draw the circles: they say the numbers and the adult writes them; or they put a card next to each pile.

If the K–1 table finishes early

Page 6 first. Then: “Find a start with six counters that takes the longest to get stuck” (**2 2 1 1**, six moves; they will probably find **4 1 1** with five). “What is the biggest staircase you can build with the whole table’s counters?” (With 50 counters: 9 steps use 45.) “Seven counters: stuck or merry-go-round?” (A merry-go-round of four: **4 2 1**, **3 3 1**, **3 2 2**, **3 2 1 1**.)

For the share: this table shows the staircases 1, 3, 6, 10 built from counters, and says two things: “six

counters always get stuck at 3, 2, 1” and “four counters never get stuck, they go round and round.”

6 Grades 2–3 table (four third graders)

The centrepiece is the **map of six**: all eleven positions, one arrow each, every road ending at **3 2 1**. It is the first time most of them will have seen a whole finite dynamical system at once. The second beat is the argument for why the staircase is stuck, and the third is the surprise that eight counters loops—in two different loops.

Order of work, with times

Minutes	Page	What happens
0:12–0:18	p. 1	The rule and notation; six counters from 6 , from 4 1 1 , and from a start of their choice. All stuck at 3 2 1 .
0:18–0:32	p. 2, p. 3	The map of six: fill the “after one move” table (share the eleven positions out: 3, 3, 3, 2), then draw the arrows on the circle. Three questions.
0:32–0:39	p. 4	Why the staircase is stuck; staircase numbers; why 8 can never be stuck.
0:39–0:48	p. 5	Eight counters: loops. Two different loops at the same table.
0:48–0:50	p. 6	Choose one challenge to take home; get ready to show the map.

If the map is still unfinished at 0:32, move on anyway and come back to it at the end: pages 4 and 5 are where the ideas are.

Answers, page by page

Page 1: warm-up

6 → **5 1** → **4 2** → **3 2 1** (3 moves). **4 1 1** → **3 3** → **2 2 2** → **3 1 1 1** → **4 2** → **3 2 1** (5 moves). Whatever start C they choose, they end at **3 2 1** within 6 moves. What to notice: everyone gets stuck, at the same position, but after different numbers of moves.

Page 2: the table

6	→	5 1	3 1 1 1	→	4 2
5 1	→	4 2	2 2 2	→	3 1 1 1
4 2	→	3 2 1	2 2 1 1	→	4 1 1
4 1 1	→	3 3	2 1 1 1 1	→	5 1
3 3	→	2 2 2	1 1 1 1 1 1	→	6
3 2 1	→	3 2 1			

Why there is no twelfth position: sort by the biggest pile. Biggest 6: only **6**. Biggest 5: the last counter is alone, **5 1**. Biggest 4: the other two counters are together or apart, **4 2**, **4 1 1**. Biggest 3: the other three make **3**, **2 1** or **1 1 1**, giving **3 3**, **3 2 1**, **3 1 1 1**. Biggest 2: **2 2 2**, **2 2 1 1**, **2 1 1 1 1**. Biggest 1: **1 1 1 1 1 1**. That is $1 + 1 + 2 + 3 + 3 + 1 = 11$.

Page 3: the map

The finished map is drawn in Appendix A (both as the circle the children have and as a “river”). (1) All roads end at **3 2 1**, which points to itself. (2) The longest road starts at **2 2 1 1**: 6 moves. (3) Three positions have no arrow pointing at them: **2 2 1 1**, **2 1 1 1 1**, **1 1 1 1 1 1**. Two positions have

two arrows pointing at them: **5 1** (from **6** and **2 1 1 1 1**) and **4 2** (from **5 1** and **3 1 1 1**); and **3 2 1** has two as well, counting its own loop. Some child will notice that the arrows “merge”; that is why there is only one ending.

Page 4: why the staircase is stuck

(1) **4 3 2 1** has four piles, so the bottom row has four counters and stands up as a new pile of 4; the old piles become 3, 2, 1 and nothing. That is **4 3 2 1** again. The same sentence works for any staircase. (2) 15, 21, 28: add 5, then 6, then 7 (the next staircase has one more, longer step). (3) No; three steps use 6 counters and the 2 left over cannot make a step of 4. (4) To be stuck you must be a staircase (Fact 2); 8 counters cannot make a staircase; so 8 counters can never be stuck. This is a complete argument, and third graders can give it.

Page 5: eight counters

Two loops. The long one: **4 3 1** → **3 3 2** → **3 2 2 1** → **4 2 1 1** → **4 3 1**. The short one: **4 2 2** → **3 3 1 1** → **4 2 2**. Fifteen of the 22 starts reach the long loop, e.g. **8** → **7 1** → **6 2** → **5 2 1** → **4 3 1**. Seven reach the short one: **5 3**, **6 1 1**, **4 1 1 1 1**, **2 2 2 2**, **2 2 1 1 1 1**, and the loop itself. E.g. **2 2 2 2** → **4 1 1 1 1** → **5 3** → **4 2 2** → **3 3 1 1**. (4) Each loop position is **3 2 1** plus two extras in two different *places*: on top of the 3, on top of the 2, on top of the 1, or as a new pile of one (Fact 4). In the long loop the two extras sit in neighbouring places; in the short loop they sit two apart, which is why it repeats after two moves.

Page 6: challenges

1. Seven counters loop through four positions: **4 2 1** → **3 3 1** → **3 2 2** → **3 2 1 1** → **4 2 1**. From 7: **6 1**, **5 2**, then **4 2 1**. 2. A 12-move road: **3 3 2 1 1** → **5 2 2 1** → **4 4 1 1** → **4 3 3** → **3 3 2 2** → **4 2 2 1 1** → **5 3 1 1** → **4 4 2** → **3 3 3 1** → **4 2 2 2** → **4 3 1 1 1** → **5 3 2** → **4 3 2 1**. Other 12-move starts: **3 2 2 2 1**, **3 2 2 1 1 1**. Roads of at least 8: any start that runs into **4 3 3** or **3 3 2 2** early, e.g. **4 3 3** itself (9), **8 1 1** (9), **5 4 1** (10), **6 2 1 1** (11). 3. **2 2 1 1**, **2 1 1 1 1**, **1 1 1 1 1 1**: many small piles—at least two more piles than the biggest pile has counters. For 7: **2 2 2 1**, **3 1 1 1 1**, **2 2 1 1 1**, **2 1 1 1 1 1**, **1 1 1 1 1 1 1**. 4. Four counters: **1 1 1 1** → **4** → **3 1** → **2 2** → **2 1 1** → **3 1**. Five counters: **1 1 1 1 1** → **5** → **4 1** → **3 2** → **2 2 1** → **3 1 1** → **3 2**, and **2 1 1 1** → **4 1**. In both cases every road ends in the same loop, but never at the same place: there is no “stuck” for 4 or 5.

If the 2–3 table stalls

- **Moves go wrong.** The usual error is forgetting the new pile or forgetting to remove a pile of one. Have them lay the piles out as columns and physically slide the bottom row out and stand it up.
- **The map takes too long.** Split the eleven positions among the four children and have each do only their three; the table on page 2 is the real content, the circle is decoration.
- **The arrows are a mess.** Fine—the river version in the appendix can be shown at the end. Ask instead: “Which position do the most arrows go into?” and “Which positions have no arrow going in?”
- **“Why is the staircase stuck” gets blank looks.** Build **3 2 1** as columns, ask “how many piles? so how many counters in the bottom row? so how big is the new pile?”, then slide the row out and stand it up at the left.
- **The loop with 8 counters is not noticed.** Ask them to write every position and to shout when they see one they have written before. If nobody finds the short loop, give the hint **5 3**.

If the 2–3 table finishes early

Page 6 in order: seven counters (a loop of four); the 12-move road (a genuine hunt—tell them the start must have five or six piles and that nobody can *reach* it, which they will recognise from the map); the unreachable positions; and the maps for 4 and 5 counters (5 and 7 positions). A further question: “*Nine counters: one loop or two?*” (One, of four positions: **4 3 2**, **3 3 2 1**, **4 2 2 1**, **4 3 1 1**.)

For the share: hold up the map of six and say: “*every road ends at 3 2 1, the longest road has 6 moves, and 8 counters never get stuck—they loop, and there are two different loops.*”

7 Grades 4–5 table (two fourth graders, one fifth grader)

This table gets the theory: which numbers get stuck (staircase numbers), a proof that only staircases are stuck, the pigeonhole argument that every game loops, the structure of the loops as “staircase plus extras”, Gardens of Eden, and the record road lengths. Pages 1–4 are the core. Page 5 (backwards) is the best material for early finishers; page 6 is a take-home. The adult should not hurry the proofs on page 3; a fifth grader who says the argument for Fact 2 in their own words has done the mathematics of the day.

Order of work, with times

Minutes	Page	What happens
0:12–0:16	p. 1	Rule, notation, warm-up from 6 and 2 2 1 1 . Read the five questions together.
0:16–0:25	p. 2	Survey 1–10 counters, split three ways (suggested: 1, 2, 3, 4 / 5, 7, 8 / 6, 9). Conjecture: the stuck numbers are the staircase numbers.
0:25–0:34	p. 3	Why: the bottom row; only staircases are stuck (fill-in); 8 is never stuck; pigeonhole.
0:34–0:44	p. 4	Loops for 7, 8, 9 as staircase plus extras; the extras march; predictions for 11 and 12, tested if time.
0:44–0:50	p. 5	Backwards: parents and Gardens of Eden. As far as they get.
—	p. 6	The longest road: for early finishers and for home.

Answers, page by page

Page 1: warm-up

6 → **5 1** → **4 2** → **3 2 1**. **2 2 1 1** → **4 1 1** → **3 3** → **2 2 2** → **3 1 1 1** → **4 2** → **3 2 1**. Same ending, different lengths (3 and 6).

Page 2: survey

counters	stuck at	or loops through
1	1	
2		2 → 1 1 → 2 (length 2)
3	2 1	
4		3 1 → 2 2 → 2 1 1 (length 3)
5		3 2 → 2 2 1 → 3 1 1 (length 3)
6	3 2 1	
7		4 2 1 → 3 3 1 → 3 2 2 → 3 2 1 1 (length 4)
8		4 3 1 → 3 3 2 → 3 2 2 1 → 4 2 1 1 (length 4), or 4 2 2 → 3 3 1 1 (length 2)
9		4 3 2 → 3 3 2 1 → 4 2 2 1 → 4 3 1 1 (length 4)
10	4 3 2 1	

(1) 1, 3, 6, 10: each is the previous plus the next counting number (1, $1 + 2$, $1 + 2 + 3$, $1 + 2 + 3 + 4$); next 15 and 21. (2) Laid out as columns each stuck position is a staircase, and a staircase with k steps has $1 + 2 + \dots + k$ counters, so only those numbers can get stuck. (3) Probably not: two loops for 8. (4) Numbers of positions: 5, 7, 11, 15 (and 22 for 8, 30 for 9, 42 for 10). Count them by biggest pile as in the 2–3 answers.

Page 3: why

(a) Fact 1: four piles, so the bottom row has four counters and becomes the new tallest column; the others shift right and shrink by one; the same shape. (b) The blanks in order: m ; new; m ; $m - 1$; $m - 1$; one; 1; staircase. Spoken version: “the new pile is the biggest, so the biggest pile equals the number of piles; every other pile after the move is an old pile minus one, so going down the list each pile is one less than the previous; with m piles that ends at 1.” The full argument is Fact 2 in §3; the fill-in is that argument with the bookkeeping hidden. (c) 8 is not a staircase number, so no position with 8 counters is a staircase, so by (b) none is stuck. (d) After 42 moves we have written down 43 positions but there are only 42, so two of them are the same (pigeonhole). The next position depends only on the current one, so from the repeat onward the game repeats the same stretch forever: it is in a loop. The same works for any number of counters, with $p(n)$ in place of 42.

Page 4: loops

7: 4 2 1, 3 3 1, 3 2 2, 3 2 1 1 (from 7 in 3 moves; from 2 2 2 1: 4 1 1 1, 4 3, 3 2 2). 9: 4 3 2, 3 3 2 1, 4 2 2 1, 4 3 1 1. 8: the two loops given in the 2–3 answers, page 5. (1) Extras: one for 7, two for 8, three for 9. The four places are: on top of the 3, on top of the 2, on top of the 1, and a new pile of one—never two extras in the same place. (2) The extra moves one place to the right each move (down the staircase); from the fourth place (the lone pile) it reappears in the first place. Explanation: Fact 4—the bottom row stands up as the new first column and everything else shifts right; the lone pile vanishes but makes the bottom row one longer, so it comes back as an extra on the new first column. (3) 4 2 2 has extras in places 1 and 3; after a move they are in 2 and 4 (3 3 1 1); after another, in 3 and 1 again. A pattern that looks the same after a half turn gives a loop of half length. (4) 11 counters: one extra in one of five places, so 5 loop positions, one loop of length 5: 5 3 2 1 → 4 4 2 1 → 4 3 3 1 → 4 3 2 2 → 4 3 2 1 1. 12 counters: two extras among five places, $\binom{5}{2} = 10$ loop positions, and since 5 is odd no pattern repeats early, so two loops of length 5: 5 4 2 1 → 4 4 3 1 → 4 3 3 2 → 4 3 2 2 1 → 5 3 2 1 1 (extras in neighbouring places) and 5 3 3 1 → 4 4 2 2 → 4 3 3 1 1 → 5 3 2 2 → 4 4 2 1 1 (extras two apart). From 12 the first loop is reached in 7 moves; from 9 3 the second, in 4.

Page 5: backwards

(1) $411 \rightarrow 33$. (2) 41 and 311 . (3) $54, 531, 4311$. (4) Impossible. Blanks: before; 2 (two); 4; 2. (5) A position is a Garden of Eden exactly when its biggest pile has fewer counters than (number of piles) $- 1$, i.e. when there are at least two more piles than the biggest pile has counters. (6) Six counters: $2211, 21111, 111111$. Seven: $31111, 2221, 22111, 211111, 1111111$. (7) 422 has 3 piles; sizes $s \geq 2$ are 4 and 2. Choosing $s = 4$: remove it, the others become 33 , add $4 - 2 = 2$ ones: parent 3311 . Choosing $s = 2$: the others become 53 , add $2 - 2 = 0$ ones: parent 53 . Both check forwards. The recipe works because a parent with s piles produces a new pile of exactly s and shrinks the rest; every parent arises this way because its new pile must be one of the piles we see, and once we know which, the rest is forced (Fact 7). (8) Yes, but not with fewer than 18 counters: four parents need four different pile sizes all at least $p - 1$, so at least four piles of sizes at least 3: 6543 , with parents $765, 7641, 75411, 654111$. With three piles at most three parents (432 is the smallest with three).

Page 6: the longest road

(1) The 12-move road from 33211 is written out in the 2–3 answers (page 6); the other 12-move starts are 32221 and 322111 . A longest start must be a Garden of Eden: otherwise its parent would have a longer road. (2) One move: 532 only. Two moves: $64, 631, 43111$. (3) $2 = 2 \cdot 1, 6 = 3 \cdot 2, 12 = 4 \cdot 3$: the record for the k -step staircase is $k(k - 1)$; for 15 counters, 20. (4) Short: $15 \rightarrow 141 \rightarrow 132 \rightarrow 1221 \rightarrow 1131 \rightarrow 1032 \rightarrow 9321 \rightarrow 8421 \rightarrow 7431 \rightarrow 6432 \rightarrow 54321$: 10 moves. (From a single pile of T_k counters the road has 1, 3, 6, 10 moves for $k = 2, 3, 4, 5$ —staircase numbers again, T_{k-1} each time. Does that continue? A nice question to leave open.) (5) A 20-move road: $443211 \rightarrow 63321 \rightarrow 55221 \rightarrow 54411 \rightarrow 5433 \rightarrow 44322 \rightarrow 533211 \rightarrow 64221 \rightarrow 55311 \rightarrow 5442 \rightarrow 44331 \rightarrow 53322 \rightarrow 542211 \rightarrow 64311 \rightarrow 5532 \rightarrow 44421 \rightarrow 53331 \rightarrow 54222 \rightarrow 543111 \rightarrow 6432 \rightarrow 54321$. Sixteen starts achieve 20. The part they cannot prove: that *every* start reaches the staircase (Fact 5); they can prove that every start loops and that the only possible stuck position is the staircase.

If the 4–5 table stalls

- **The survey drags.** Three starts per number is plenty; the point is the conjecture. Skip 1 and 2 if needed (1 is stuck trivially; 2 flips between 2 and 11).
- **The fill-in argument (b) does not land.** Build a supposed stuck position from counters and ask: “after the move, where is the new pile in the list? Could it be anything but the biggest? Now cover the biggest pile on both sides—what is left?” Accept an argument in words; the blanks are scaffolding.
- **They do not see the extras.** Build 321 in one colour and put the loop position’s extras in the second colour (coins). Then do a move with the coins visible: the coins walk.
- **Page 5 (4) frustrates.** Good; let them try for two minutes, then ask “how big must the new pile be if there were four piles before?”

If the 4–5 table finishes early

Page 6, then: “Find every position with 8 counters that has no parent” (seven: $32111, 311111, 2222, 22211, 221111, 2111111, 11111111$). “Draw the whole map for 7 counters” (15 positions, one loop of four; every road reaches the loop within 4 moves). “How many loops for 13 and 14 counters?” (Two of length 5, then one of length 5.) “Is the road from a single pile of T_k always a staircase number of moves?” (For $k \leq 7$ it is T_{k-1} ; let them test $k = 6$ at home: 21 takes 15 moves.)

For the share: one child explains why only a staircase can be stuck (Fact 2, in words); another shows the two loops for 8 counters as the staircase plus two extras; a third names a Garden of Eden and says why nothing leads to it.

8 The share and the closing (0:50–1:00)

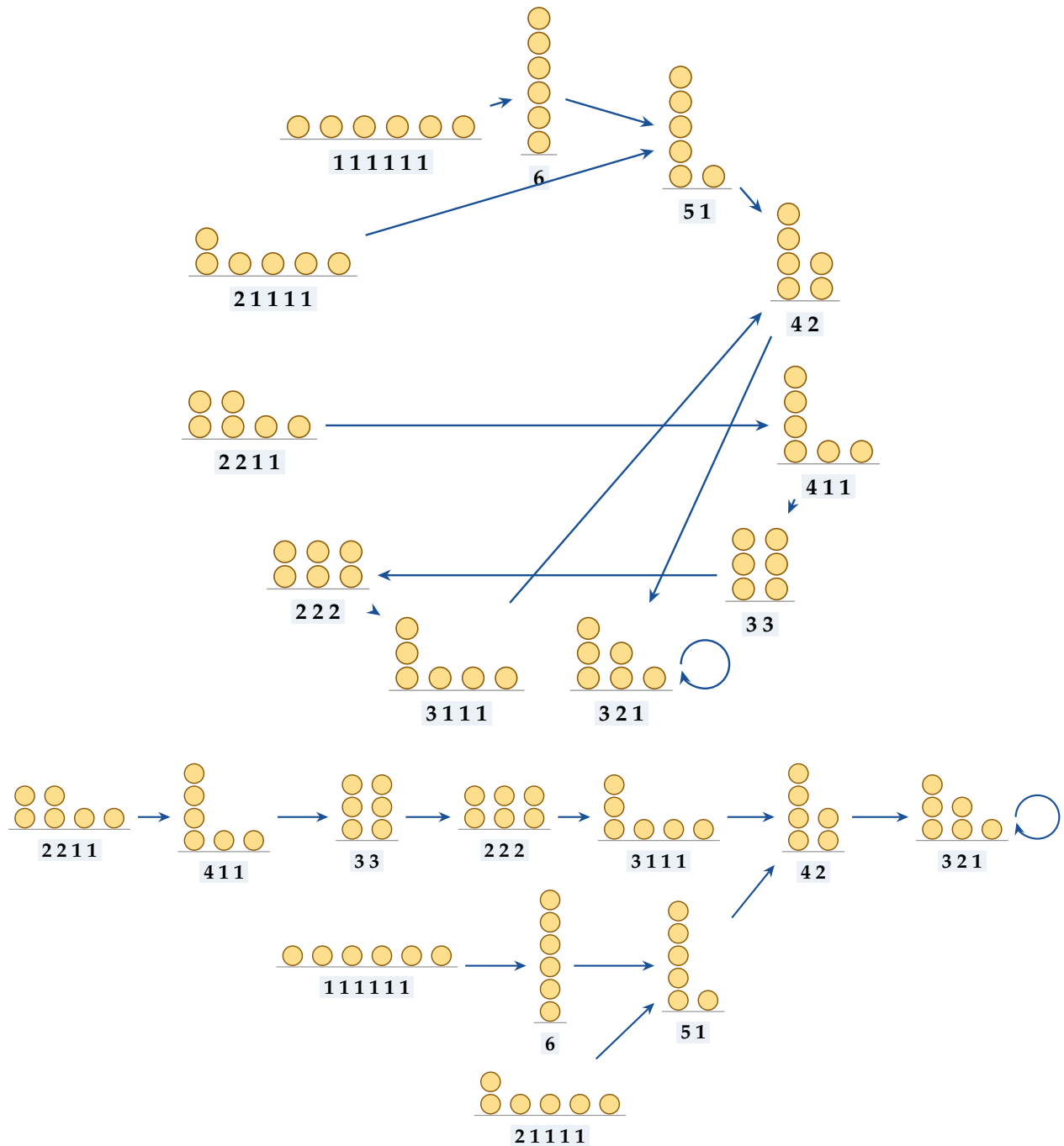
Youngest table first, two to three minutes each; the organizer keeps time and asks one question per table.

- **K–1:** the staircases 1, 3, 6, 10 on the table; *“six counters always get stuck at 3, 2, 1; four counters never get stuck.”* Organizer’s question: *“what did ten counters do?”*
- **Grades 2–3:** the map of six held up; the longest road; *“8 counters can’t get stuck, and there are two different loops.”* Question: *“why can’t 8 get stuck?”* (Only staircases are stuck, and 8 is not a staircase number.)
- **Grades 4–5:** the argument that only staircases are stuck; the loops as staircase-plus-extras; a Garden of Eden. Question: *“how do you know every game ends in a loop?”*

“Here is what you found today, and it is a theorem: if the number of counters is a staircase number, every start ends stuck at the staircase. A mathematician named Brandt proved it in 1982, and people are still studying this game—for instance, exactly how long the longest road can be: with 15 counters it is 20 moves. Take-home question: find a start with 15 counters that takes 20 moves. And for the youngest: find out what 7 counters do.”

A Data

The map of six



Moves to the staircase from each start: **321** 0; **42** 1; **51**, **3111** 2; **6**, **222**, **21111** 3; **33**, **111111** 4; **411** 5; **2211** 6.

Summary for 1 to 15 counters

counters	positions	loop lengths	in loops	longest road	Gardens of Eden	a longest start
1	1	1 (stuck)	1	0	0	1
2	2	2	2	0	0	
3	3	1 (stuck)	1	2	1	1 1 1
4	5	3	3	2	1	1 1 1 1
5	7	3	3	3	2	1 1 1 1 1
6	11	1 (stuck)	1	6	3	2 2 1 1
7	15	4	4	4	5	1 1 1 1 1 1 1
8	22	4, 2	6	5	7	1 1 1 1 1 1 1 1
9	30	4	4	7	10	3 2 2 1 1
10	42	1 (stuck)	1	12	14	3 3 2 1 1
11	56	5	5	8	20	3 3 2 2 1
12	77	5, 5	10	8	27	3 3 2 2 1 1
13	101	5, 5	10	9	37	1 1 1 1 1 1 1 1 1 1 1 1 1 1
14	135	5	5	14	49	4 3 3 2 1 1
15	176	1 (stuck)	1	20	66	4 4 3 2 1 1

“Longest road” is the largest number of moves from any start until the game is stuck or has entered its loop. For $n = T_k + r$ the positions in loops are the $\binom{k+1}{r}$ staircase-plus-extras positions of Fact 4.

All loops for 7 to 14 counters

- 7: 4 2 1 → 3 3 1 → 3 2 2 → 3 2 1 1 → 4 2 1.
- 8: 4 3 1 → 3 3 2 → 3 2 2 1 → 4 2 1 1 → 4 3 1; 4 2 2 → 3 3 1 1 → 4 2 2.
- 9: 4 3 2 → 3 3 2 1 → 4 2 2 1 → 4 3 1 1 → 4 3 2.
- 11: 5 3 2 1 → 4 4 2 1 → 4 3 3 1 → 4 3 2 2 → 4 3 2 1 1 → 5 3 2 1.
- 12: 5 4 2 1 → 4 4 3 1 → 4 3 3 2 → 4 3 2 2 1 → 5 3 2 1 1 → 5 4 2 1; 5 3 3 1 → 4 4 2 2 → 4 3 3 1 1 → 5 3 2 2 → 4 4 2 1 1 → 5 3 3 1.
- 13: 5 4 3 1 → 4 4 3 2 → 4 3 3 2 1 → 5 3 2 2 1 → 5 4 2 1 1 → 5 4 3 1; 5 4 2 2 → 4 4 3 1 1 → 5 3 3 2 → 4 4 2 2 1 → 5 3 3 1 1 → 5 4 2 2.
- 14: 5 4 3 2 → 4 4 3 2 1 → 5 3 3 2 1 → 5 4 2 2 1 → 5 4 3 1 1 → 5 4 3 2.

Some useful roads

- Ten counters from 10: 9 1, 8 2, 7 2 1, 6 3 1, 5 3 2, 4 3 2 1 (6 moves). From 1 1 1 1 1 1 1 1 1 1: 10, then as above (7 moves). From 2 2 2 2 2: 5 1 1 1 1 1, 6 4, 5 3 2, 4 3 2 1 (4 moves). From 4 3 3: 3 3 2 2, 4 2 2 1 1, 5 3 1 1, 4 4 2, 3 3 3 1, 4 2 2 2, 4 3 1 1 1, 5 3 2, 4 3 2 1 (9 moves).
- Eight counters from 1 1 1 1 1 1 1 1: 8, 7 1, 6 2, 5 2 1, 4 3 1 and the long loop (5 moves, the longest for 8).
- Nine counters from 9: 8 1, 7 2, 6 2 1, 5 3 1, 4 3 2 and the loop (5 moves). From 3 2 2 1 1: 5 2 1 1, 4 4 1, 3 3 3, 3 2 2 2, 4 2 1 1 1, 5 3 1, 4 3 2 (7 moves, the longest for 9).
- Seven counters from 1 1 1 1 1 1 1: 7, 6 1, 5 2, 4 2 1 and the loop (4 moves, the longest for 7).

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Every number and road in this guide was computed by exhaustive enumeration of all positions up to 21 counters.